

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester M.Tech Degree Regular/PhD Even Semester (M.Tech New Scheme) Examination June 2023

Discipline: COMPUTER SCIENCE AND ENGINEERING**Course Code & Name: 222TCS002 DEEP LEARNING**

Max. Marks: 60

Duration: 2.5 Hours

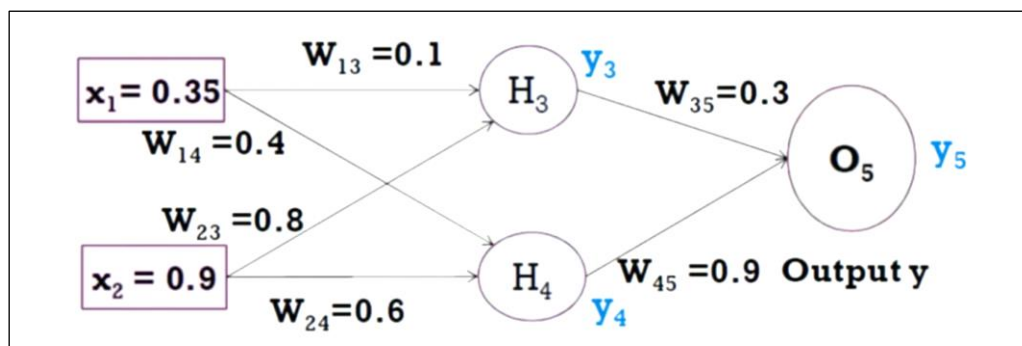
PART A*Answer all questions. Each question carries 5 marks*

Marks

- | | | |
|---|--|-----|
| 1 | Explain the different types of Recurrent Neural Networks with diagrams | (5) |
| 2 | Analyze the Advantages of Convolutional Neural Networks (CNNs) Compared to Artificial Neural Networks (ANNs) for Image Data Input. | (5) |
| 3 | Analyze the impact of the learning rate on the training process. | (5) |
| 4 | Evaluate the effectiveness of BERT (Bidirectional Encoder Representations from Transformers) in the context of automated question answering. | (5) |
| 5 | Identify challenges associated with training and implementing Transformers | (5) |

PART B*Answer any 5 questions. Each question carries 7 marks*

- | | | |
|---|---|-----|
| 6 | Perform a forward pass and a backward pass on the provided multilayer perceptron network. Assume a sigmoid activation function for the neurons. Given an actual output of 0.5 and a learning rate of 1, carry out another forward pass. Apply the backpropagation algorithm to update the network's weights. Finally, calculate the error using appropriate error calculation techniques. | (7) |
|---|---|-----|



- 7 Illustrate and compare the concepts of batch gradients and stochastic gradients in the context of optimization algorithms for deep learning. (7)
- 8 Explain the different layers of convolution neural networks with the help of diagrams (7)
- 9 Examine the idea behind Recurrent neural networks and the vanishing gradient problem in RNN and name a solution for the same (7)
- 10 Demonstrate the workflow of the BERT technique of word embedding. (7)
- 11 Explain the key components of the Transformer architecture, including encoders, decoders, self-attention mechanisms, working of transformers, and application areas in Natural language processing (7)
- 12 Illustrate the architecture of a Vision Transformer (ViT), highlighting its key components and their interactions. (7)
